UNIT 3:

FOOD SAFETY STANDARDS

Q: List out the important food safety standards and regulations.

FSSAI (Food Safety and Standards Authority of India):

Q: Expand the term FSSAI, illustrate its functions, importance & type of license issued by them.

A:

The full form of FSSAI is the Food Safety and Standards Authority of India. The organization belongs to the Ministry of Health and Family Welfare of the Indian Government. To ensure that healthy and nutritious foods are accessible to humans, it aims to establish scientific standards for the control, inspection and regulation of the production, storage, selling, distribution and import of foods.

Functions:

- Framing of laws to set down food safety standards.
- Authorizing standards to accredit labs for food testing.
- Providing the Central Government with technical assistance and scientific guidance.
- Conducting surveys to obtain food intake and food pollution information.
- Spread data to raise awareness regarding food quality.
- Conduct training programmes to educate individuals who have been in or want to begin a food business.

Importance of FSSAI

- Under the Food and Safety Act of 2006, FSSAI was established.
- Its headquarter is located in New Delhi, India.
- It has 8 regional offices which are located in Chandigarh, Delhi, Lucknow, Mumbai, Guwahati, Kolkata, Cochin & Chennai.
- There are 4 referral laboratories and 72 local food testing labs for the testing of adulterated products.
- The organization has the power to perform chemical research on various firms' food items.
- It also makes progress against businesses that do not ensure the appropriate nutrition & quality of the food items. For example, FSSAI banned Maggi noodles.
- FSSAI is allowed to grant licenses that conform to its rules and practices to Food Business Operators.The license is now compulsory for anyone who is in a food company, like a food processor, food trader and owner of a hotel or restaurant.

The three licenses are –

- **FSSAI Basic registration :** Small businesses or startups having annual turnover below Rs.12 lakhs can apply. As operations scale up and turnover reaches Rs. 12 lakh bar, the basic registration will need to be upgraded to state license.
- **FSSAI State license** : Small and Medium business units having annual turnover in the range of Rs.12 lakh to 20 crores need to apply for State level License.
- **FSSAI Central license** : For large businesses having annual turnover above Rs.20 crores, applicants need to obtain Central Level License. The same license is applicable where you are engaged in import or export of food products or selling goods to government offices.

Procedure Involved in FSSAI Testing :

1. Signature of Witnesses : The Food Safety officer should call for one or more witnesses while taking the sample piece and obtain a signature on all related forms and documents.
2. Notice to the Business Operator : If the product has been obtained from the manufacturer or supplier, a notice in Form V-A shall be issued to them. When a product is drawn from an open container, the person drawing the sample should draw another sample from a container in its original condition bearing the same declaration and intimate the same to the Food Analyst.
3. Payment for the Sample : The Food Safety Officer should pay the cost of the sample to the person from whom the sample drew it. The price should be calculated at the rate at which the product is sold to the public.
4. Packing of Sample : The sample food for the analysis should be taken in clean, dry bottles or jars or in a container that should be adequately sealed to prevent leakage, and evaporation and to avoid the entrance of moisture. When selecting an already packed product as a sample, no further sealing is required.

5. Labeling the Package : The package should be properly labeled and should bear the following requirements:
    a. Code of the sample
    b. Name of the sender with his official designation
    c. Date and place of collection
    d. Nature of the product that has been sent for analysis
    e. Nature and quantity of preservative added to the sample
6. Sealing of package : The sample should be divided into four parts, and each piece should be sealed and wrapped using strong paper. Further, a paper slip that covers the bottom to the top of the container, bearing the signature of the designated officer, should be pasted on the wrapper. The paper cover should be further protected by means of a strong thread above and across the jar. The knots of the thread should be covered by wax bearing the impression of the sender's seal.
7. Dispatch of the Package : The containers of the samples shall be dispatched in the following manner:
    a. The sealed container of one part of the sample, along with the memorandum in Form VI should be sent to the Food Analyst.
    b. The second and third parts of the sample along with two copies of the memorandum in Form VI should be sent to the Designated Officer.
    c. The fourth part of the sample along with a copy of the memorandum in Form VI should be sent to an accredited laboratory along with the prescribed fee.

On receipt of the food sample, the authorized FSSAI lab would analyze the quality and safety of the sample food product and pass the final decision.

Compliance Rules :

FSSAI compliance refers to adhering to the regulations and guidelines set by FSSAI to ensure that food products are safe and of good quality for consumption. FSSAI compliance involves obtaining a food license, adhering to food safety and labeling standards, maintaining proper hygiene and sanitation in food processing and handling, and undergoing regular inspections and audits to ensure compliance with FSSAI regulations. Compliance with FSSAI regulations is essential for food businesses in India to operate legally and ensure the safety and well-being of their consumers.

Offenses :

1. Selling substandard or misbranded food products
2. Manufacturing or selling food products that are unsafe for consumption
3. Failing to obtain a food license
4. Failing to comply with food safety and labeling standards
5. Failing to implement GMP and GHP
6. Adulteration of food products
7. Obstructing an FSSAI official from performing their duties: FBOs who obstruct an FSSAI

FOOD AND HYGIENE

Microbial Food Spoilage and Food Borne Diseases:

Food spoilage is a metabolic process that causes foods to be undesirable or unacceptable for human consumption due to changes in sensory characteristics.

- identified by off-colors, off-odors, softening of vegetables, fruits, and slime production.
- arise from insect damage, physical damage, and indigenous enzyme activity in food or by microorganisms (bacteria, viruses, fungi).
- The spoilage microbe's common inhabitants are soil, water, or the intestinal tracts of animals or they are dispersed through the air and water.
- depends upon intrinsic (pH, water activity, nutrient content, oxidation-reduction potential, antimicrobial property) and extrinsic factors (temperature, relative humidity, pressure).

Q: List out the food spoilage microorganisms and explain the diseases caused by them.

A:

Foodborne illnesses may result from any of the following sources:

- Bacteria: These may be present in raw and undercooked meat, fish, and poultry; unpasteurized dairy products; contaminated fruits and vegetables; and contaminated drinking water.
- Viruses: Viruses are transmitted to the body through food contaminated by viral particles.
- Parasites: Contaminated water and soil can transmit harmful parasites to fresh produce, seafood, meat, poultry, and other foods.
- Prions: These infectious proteins are associated with "mad cow disease" and can come from eating parts of cattle, such as the brain tissue.
- Naturally occurring chemicals: Naturally occurring toxins in mushrooms, staple foods such as corn and cereal, and mold on grains can cause long-term health complications.
- Environmental pollutants: Byproducts of plastic production and waste management, as well as heavy metals such as lead and mercury found in water and soil, can contaminate foods and lead to foodborne illnesses.

Organism	Common Name of Illness
Hepatitis A	Hepatitis
Listeria monocytogenes	Listeriosis
Noroviruses	Variously called viral gastroenteritis, winter diarrhea, acute non- bacterial gastroenteritis, food poisoning, and food infection
Salmonella	Salmonellosis

Q: What is Food Hygiene?Mention the physical, chemical and biological hazards which contaminate food. List out various methods to eliminate those from food.

A:

Food Hygiene: the practice of handling, preparing, and storing food in a way that prevents foodborne illnesses.

Physical Hazards : Physical contamination occurs when a physical object enters food at some stage of the production or preparation process.

Ex: Hair, Fingernails, Broken glass, staples, Plastic wrap/packaging, Dirt from unwashed fruit and vegetables, Pests, pest droppings, rodent hair

To eliminate :

- Cleanliness … Personal hygiene
- Use of Magnetic separator to separate metallic items
- Maintenance of equipments, buildings, utilities or instruments
- Use of Hair nets, masks and gloves
- No jewelry or minimum
- Wash fruits and vegetables thoroughly

Chemical Hazards : Chemical contamination occurs when food produces or comes into contact with toxic chemicals, which can lead to chemical food poisoning. Chemical contaminants fall into one of two categories: natural and artificial.

Naturally-occurring toxins are toxic compounds that are produced by living organisms, some of which are staples of the human diet (e.g. shellfish, potatoes, fish). These toxins are not harmful to the organisms themselves but can be harmful to us if we eat them. Artificial toxins include Cleaning products (e.g. detergent, sanitizer), Pesticides/herbicides, Toxic chemicals in metals and plastics.

To eliminate :

- Cleaning, washing and sanitizing
- Check quality of water before using it for food or consumption
- GAP
- Label and store chemicals separately from food
- Ventilation
- Use of Air curtains

Biological Hazards : *//write about bacteria , fungi… given in microbial food spoilage and mention insects and rodents*

To eliminate :

- Pest control, Fly killers and Rat baits
- Consume food within the required time from preparation.Food storage :
    - Less than 2 hr : may be stored more
    - 2 to 4 hours : cooked, consumed or discarded
    - More than 4 hours : must be discarded
- Maintaining proper temperature to prevent growth of microbes
- Maintain low Moisture , acidity and oxygen level in food

Hazards and control measures

Hazard	Control measures
Biological	Raw material: Analysis of raw material ,maturity indices, purchase specifications; pH/hardness/chemical residues/ temp. of processing water Operations: Concentration/strength of disinfectant and chemicals; washing time, temperature of water, intact packing, storage temperature
Physical	Proper use of magnets, visual inspection, preventive maintenance of equipment
Chemical	Use of food grade chemicals, compliance to GMP, separate storage for hazardous substance, purchase specifications, surveillance program for raw materials, Raw material

Food Analysis and Testing

Food testing is crucial for food industries for many reasons as it covers wide range of requirements like

- Testing the quality of a product
- Quality control
- Food inspection and grading
- Food authenticity check
- Testing for adulterants
- Nutritional labeling

- Food nutrient content analysis
- Shelf-life analysis
- Sensory evaluation

Food Testing can benefit food production business

- Customers: Protects and guaranteed prevention to customers from food-borne diseases.
- Manufacturers: Assures product safety, approval to enter the market, and aids in brand reputation which in turn, leads to an increase in business ventures.

Q: Techniques that take place in a food testing laboratory

A:

1. **Analytical chemistry testing:** This helps to identify, separate and even find the quantity of the chemical components of natural and artificial materials in the food product such as pH, additives, nutrients, preservatives, contaminants, minerals and much more.
2. **Microbiology testing:** This food testing helps food manufacturers identify the microorganisms that inhabit the food and understand the safety of the raw materials used in food production, ingredients as well as the final product. This is also used to test and examine spoiled pathogens to prevent food poisoning outbreaks that are generally caused by the products and the ingredients used in the products. This is also a necessity otherwise the whole supply chain can be contaminated and disrupted in the process of food production, leading to not only a danger to the general public but also a hefty loss of money.
3. **Food allergen test:** Food allergens are proteins that appear in huge quantities. It helps to find the main target allergen in the ingredients and finished food product. Normally allergens are tested in food products such as peanuts, gluten in grains, eggs, nuts and milk, etc.
4. **Nutritional analysis:** This area in food testing provides accurate information on the nutritional content of food products. This gives manufacturers the required details to include it on food packaging and ensure that they are following all compliances and labeling regulations.

Q: 5 Methods for Food Analysis

A:

1. **Mass spectrometry (MS) :** The technique is often paired with liquid (LC) or gas chromatography (GC) to enhance results. Depending on the application, it can also be coupled with capillary electrophoresis (CE) and infrared spectroscopy (IR).
2. **Nuclear magnetic resonance (NMR) spectroscopy :** NMR spectroscopy uses the magnetic properties of atomic nuclei, usually hydrogen, carbon-13, phosphorus-31 and deuterium, to analyze liquid and solid samples. As a non-destructive method, it's ideal for analyzing complex food samples and offers the scope to simultaneously detect and

quantify multiple compounds. Scientists also covet NMR for its versatility and high reproducibility.

3. **Polymerase chain reaction (PCR):**While PCR testing has become a widely recognised term in the wake of the COVID-19 pandemic, food analysis scientists have been relying on the method for decades. The technique analyzes specific DNA sequences and has proved especially useful for identifying animal species in meat products. This helps to prevent food fraud and avoid situations. PCR testing is also used to detect harmful pathogens and other microorganisms, as well as the presence of genetically modified organisms (GMO).
4. **Enzyme-linked immunosorbent assay (ELISA):** From product manufacturers to quality control agencies, ELISA is one of the most widely used methods in the food industry. Tests are used to detect hidden allergenic proteins in food products, including hormones, antibodies and peptides. As a highly-sensitive technique, ELISA is also used to detect peanut traces in raw, processed and cooked foods.In the dairy industry, ELISA is used to identify pathogens such as E. coli, Salmonella and listeria in milk and cheese products. The testing method is also capable of detecting mold spores before they begin to grow.
5. **LFD (lateral flow device):** In the food industry, this technique is a fast and efficient way to detect gluten in products, with an antigen-antibody reaction used to identify the protein. The latest LFD testing kits are compact and easy to use, making them ideal for on-site testing.

HACCP (Hazard Analysis Critical Control Point) :

HACCP (Hazard Analysis Critical Control Point) is defined as a management system in which food safety is addressed through the analysis and control of biological, chemical, and physical hazards from raw material production, procurement and handling, to manufacturing, distribution and consumption of the finished product. The goal of HACCP is to prevent and reduce the occurrence of food safety hazards.

Q: Seven Principles of HACCP :

A:

1. **Conduct a hazard analysis :** involves listing the steps in the process and identifying where significant hazards are likely to occur. The HACCP team will focus on hazards that can be prevented, eliminated or controlled by the HACCP plan. A justification for including or excluding the hazard is reported and possible control measures are identified.
2. **Determine the critical control points (CCPs) :** A critical control point (CCP) is a point, step or procedure at which control can be applied and a food safety hazard can be prevented, eliminated or reduced to acceptable levels. The HACCP team will use a CCP decision tree to help identify the critical control points in the process. The number of

CCP's needed depends on the processing steps and the control needed to assure food safety.

3. **Establish critical limit(s) :** A critical limit (CL) is the maximum and/or minimum value to which a biological, chemical, or physical parameter must be controlled at a CCP to prevent, eliminate, or reduce to an acceptable level the occurrence of a food safety hazard. The critical limit is usually a measure such as time, temperature, water activity (aw), pH, weight, or some other measure that is based on scientific literature and/or regulatory standards.
4. **Establish a system to monitor control of the CCP :** The HACCP team will describe monitoring procedures for the measurement of the critical limit at each critical control point. Monitoring procedures should describe how the measurement will be taken, when the measurement is taken, who is responsible for the measurement and how frequently the measurement is taken during production.
5. **Establish the corrective action:** Corrective actions are the procedures that are followed when a deviation in a critical limit occurs. The HACCP team will identify the steps that will be taken to prevent potentially hazardous food from entering the food chain and the steps that are needed to correct the process. This usually includes identification of the problems and the steps taken to assure that the problem will not occur again.
6. **Establish procedures for verification:** Those activities, other than monitoring, that determine the validity of the HACCP plan and that the system is operating according to the plan. The HACCP team may identify activities such as auditing of CCP's, record review, prior shipment review, instrument calibration and product testing as part of the verification activities.
7. **Establish Record-keeping and Documentation Procedures :** A key component of the HACCP plan is recording information that can be used to prove that the food was produced safely. The records also need to include information about the HACCP plan. Record should include information on the HACCP Team, product description, flow diagrams, the hazard analysis, the CCP's identified, Critical Limits, Monitoring System, Corrective Actions, Recordkeeping Procedures, and Verification Procedures.

Planning and Realization of Safe Products :

1. General planning and realization of safe products : Process planning , Production / processing / service activities
2. Prerequisite programmes (PRPs) :
    - GAP, GAHP, GSP, GHP
    - Design and layout of processing unit
    - Maintenance, cleaning, disinfection
    - Contact surfaces – non-toxic, easy to clean
    - Adequate workspace
    - Employee facilities
    - Temperature and humidity control
    - Pest control systems
    - Personnel hygiene, GHP

    - Handling and transportation
    - Recall procedures
    - Customer complaints - handling
3. Preliminary steps to enable Hazard Analysis
    1. Establish a team (Food technologist / food nutritionist / process engineer /microbiologist / biochemist)
    2. Describe product characteristics (raw and final)
    3. Identify the products intended use (fresh-cut fruits –direct consumption; FC vegetables – cooking; ready to eat)
    4. Prepare flow chart / diagram
    5. On-site confirmation of flow diagram
    6. Identify the hazards, conduct hazard analysis and determine control measures - Biological Hazard, Chemical Hazards, Physical Hazards
4. Hazard Analysis
5. Establishing the oPRPs
    - Hazards to be controlled by oPRPs
    - Control measures
    - Monitoring procedures
    - Corrections and corrective actions
    - Responsibilities & authorities
    - Record monitoring
6. Establishing the HACCP Plan
    - Identification of CCPs
    - Determination of critical limits for CCPs
    - System for the monitoring of CCPs
    - Actions when monitoring results exceed critical limits
7. Updating of preliminary information and specifying the PRPs and the HACCP plan
    - Product characteristics
    - Intended use
    - Flow diagrams
    - Process steps
    - Control measures
8. Verification planning
    - Equipment Calibration
    - Review of Purchase specification
    - Soil sampling analysis
    - Determination and observations of OPRPs and PRPs
    - Monitoring of CCPs and critical limits
    - Training for staffs and skilled operators
9. Traceability system …. Keeping records
    - All operational manuals including GMP, GAP, CCP.
    - Overall assessment reports for:
        i. Temperature logs
        ii. Hygiene Records

iii. Receiving inspection records
iv. Medical Reports of the workers
v. Equipment calibration records
vi. Transportation records.

10. Control of nonconformity
    - Corrections
    - Corrective actions
    - Handling of potentially unsafe products
        i. Evaluation for release
        ii. Disposition of nonconforming products]
    - Withdrawal

Quiz:

1. In food, the term used for disease causing microorganism is referred to as **Pathogen.**
2. Mention the temperature range which is optimum for bacterial growth in food : **5 to 60 degree C**
3. Mention the instrument used to multiply the DNA : **RTPCR**
4. What is the 6th principle of HACCP ? : **Establish verification procedure**
5. Mention the allowable concentration of the lead in food as per the FSSAI : **2.5 ppm**
6. Mention any two-food certification followed in the world: **Halal, Kosher**
7. Mention the number of Licenses issued by FSSAI : **3**
8. Mention any two examples for microbiological contamination : **Coliform,pseudomonas**
9. Cholera is caused by **bacteria** microorganism.
10. What is Food borne disease? Give examples.
11.